

IGIS
INSTITUTE OF GEOGRAPHICAL INFORMATION SYSTEMS

INTERNATIONAL TRAINING WORKSHOP

Remote Sensing *for Disaster Management*

From satellite constellations to UAV systems: an integrated observation chain for emergency response and loss assessment

18–20 November 2026
IGIS, SCEE, NUST · Islamabad

Programme booklet

Two scales. One observation chain.

During a monsoon flood, responders need two answers at once: *how far has the water spread, and who is affected?* Then: *which embankments breached, which bridges failed, how many homes are lost?*

> No single sensing platform can address both. Used in a deliberate sequence, satellites and UAVs create an observation chain far more effective than either alone.

Over two and a half days you will work through that chain, using openly licensed satellite data, browser-based cloud computing and standard commercial UAVs, all reproducible with equipment ordinarily available to a Pakistani institution.

BY THE CLOSE OF DAY 3 YOU WILL HAVE PRODUCED

1. **A flight plan** for a real survey mission
2. **Drone imagery**: orthophoto, oblique and thermal
3. **An orthophoto map** from a survey you helped fly
4. **A 3D model** from oblique photography
5. **A UAV response plan** from the capstone exercise

PROGRAMME AT A GLANCE

From the wide view to the close-up.

01

WEDNESDAY 18 NOVEMBER · 09:15–16:00

UAV and remote sensing basics

Modules 1–4 · UAVs, remote sensing, disaster applications and sensors · flight planning exercise

02

THURSDAY 19 NOVEMBER · 09:15–16:00

UAV data processing and practice

Module 5 data processing · mission flights at the NUST field site · processing software practical

03

FRIDAY 20 NOVEMBER · 09:15–12:15, CLOSING 14:30

Integration, case studies & rapid response

Module 6 · three Pakistani case studies · capstone simulation · closing ceremony after Friday prayers

2½	14	30	4
days	contact hours	participants	expert lecturers

DAY 1 · WEDNESDAY 18 NOVEMBER

UAV and remote sensing basics

MORNING

09:00	**Registration** Badge collection and laptop check.
09:15	**Opening ceremony** Welcome address and inauguration of the workshop.
09:35	**Module 1 · Introduction to UAVs** Definition, development and characteristics; system composition and principles; norms, regulations and operating procedures; choosing a remote sensing platform by flight time, altitude, speed, payload and integration.
10:35	Tea break
10:50	**Module 2 · Remote sensing basics** Remote sensing types and scope; fundamentals of photogrammetry; applications; software demonstration.
11:50	**Module 3 · UAV remote sensing in disaster management** Risk assessment, monitoring and early warning, emergency response, and loss assessment and recovery, through practical cases; advantages over traditional methods.
12:30	Lunch and prayers

Sensors and flight planning

AFTERNOON

13:45	**Module 4 · Sensors and data acquisition**	How optical, multispectral, thermal and LiDAR sensors work; geometric correction and radiometric calibration; obtaining high-quality, usable data.
14:45	Tea break	
15:00	**Flight planning exercise**	Delineating the flight area and avoiding restricted zones; altitude, speed and overlap for the sensor and terrain; automatic and manual route planning. Groups plan the mission they fly on Day 2.
15:45	**Wrap-up and Q&A**	Recap and a preview of Day 2.
16:00	Close of Day 1	

DAY 2 · THURSDAY 19 NOVEMBER

UAV data processing and practice

09:00	Arrival and tea
09:15	**Module 5 · UAV data processing** Data products and processing workflow; POS data; image preprocessing, quick stitching, registration and correction; oblique photography and mainstream 3D software, from import to modelling and output.
10:30	Transfer to the NUST field site Near the Saddle Club.
10:45	**Practical 1 · Mission flights** Basic operations; emergency take-off and landing; one-click return-to-home; flying the planned route; orthophoto, oblique and thermal data collection.
12:30	Return to IGIS, lunch and prayers
13:45	**Practical 2 · Processing software** Hands-on preprocessing, orthophoto generation and oblique 3D modelling on IGIS workstations.
15:15	Tea break
15:30	**Wrap-up and Q&A** Recap and a preview of Day 3. Close at 16:00.

A prepared dataset keeps the processing practical on time and is the backup if weather prevents flying.

DAY 3 · FRIDAY 20 NOVEMBER

Integration, case studies & rapid response

09:00	Arrival and tea	
09:15	**Module 6 · UAVs in a real disaster response**	Deciding where to fly, getting permissions quickly, turning imagery into products responders can use, and who does what.
09:45	**Applied case studies**	Attabad Lake flood risk · Shisper Lake outburst flood scenarios · Jhang riverine flooding and the monsoon.
10:20	Short break	Scenario packs handed out.
10:30	**Capstone · Rapid-response simulation**	Groups take a disaster scenario and prepare a short UAV response plan: what to survey and why, aircraft and sensors, a flight plan, and the products they would deliver.
11:30	**Group presentations**	Each group presents, with feedback from the instructors.
12:15	Friday prayers and lunch	
14:30	**Closing ceremony**	Course summary, remarks by dignitaries, certificates and group photograph.

LEARNING OUTCOMES

What you'll be able to do.

01 Explain UAV system composition and types, and choose a platform for a task.

02 Follow Pakistan's UAV regulations, permissions and flight-safety practice.

03 Describe remote sensing types and the fundamentals of photogrammetry.

04 Explain how UAV remote sensing supports each stage of disaster management.

05 Describe how optical, multispectral, thermal and LiDAR sensors work.

06 Plan a UAV survey: area, altitude, speed, overlap and route.

07 Fly a supervised mission and collect orthophoto, oblique and thermal data.

08 Process UAV imagery into an orthophoto and an oblique 3D model.

09 Prepare a UAV response plan for a disaster scenario.

EXPERT LECTURERS

Your lecturers.



Dr. NIE Yong

Professor · Institute of Mountain Hazards and Environment (IMHE), CAS

Director, Mountain Science Data Center. Glacier and glacial lake hazards and remote sensing; 50+ papers including ESI highly cited work; risk assessment for the China–Pakistan Economic Corridor.



Engr. WANG Wen

Assistant Engineer · Mountain Science Data Center, IMHE, CAS

Remote sensing for mountain hazards and the cryosphere; extensive UAV fieldwork and processing, from flight planning to orthomosaics, DEMs and 3D models.



Dr. Javed Iqbal

Professor · SCEE – IGIS, NUST

Three decades in GIS, remote sensing and hazard mapping of landslides, glacier avalanches and flash floods; 200+ scholarly contributions.



Dr. Shahid Nawaz Khan

Assistant Professor · SCEE – IGIS, NUST

Explainable machine learning for Earth observation, SAR–optical fusion, Earth Engine and UAV remote sensing; FAA Part 107 remote pilot.

BEFORE YOU ARRIVE

Come ready to work.

The schedule is compressed, so please have these ready before Day 1.

- [ ] **A laptop** for demonstrations and exercises.
- [ ] **QGIS 3.34 LTR** installed, with 5 GB free disk space.
- [ ] **UAV ground-station app** on a charged phone or tablet, with a cable.
- [ ] **Field clothing for Day 2:** closed-toe shoes, sun protection and a hat.
- [ ] **Pre-reading pack**, circulated in advance.
- [ ] **Signed field-safety acknowledgement**, handed in at registration.

VENUE & CONTACT

Where to find us.

LECTURES & LABS · ALL THREE DAYS

Institute of Geographical Information Systems

SCEE, NUST Sector H-12, Islamabad 44000

UAV FLIGHTS · DAY 2 MORNING

NUST Field Site

Near the Saddle Club, NUST H-12 · transfer from IGIS at 10:30

NUST_CPJRC_WRKSHP@igis.nust.edu.pk
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NOTES

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IGIS
INSTITUTE OF GEOGRAPHICAL INFORMATION SYSTEMS



China-Pakistan Joint Research Center on Earth Sciences (CPJRC)
Institute of Geographical Information Systems, School of Civil and Environmental Engineering, NUST
Geological Survey of Pakistan

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